

Assignment - 1

Part A

1. Full stack Development is the process of developing both the frontend and backend of a web application.

A Full Stack Developer is a professional who can work on:

Frontend, Backend, Database, APIs

ex- An e-commerce website where the developer builds the product page, manages user accounts and stores data in a database.

2. Frontend is the part of a website or application that users can see and interact with directly.

Technologies used -

HTML, CSS, JS, React, Angular, Vue.js

ex- Buttons, menus, forms, images and web pages layouts.

3. Backend is the server-side part of a website that processes requests, performs business logic, and communicates with databases.

Main Responsibilities:-

Processing user requests.

Authentication and authorization

Data management

Communication with databases

ex- Verifying login credentials when a user signs in

4. A database is an organized collection of data stored electronically. Websites need databases to store and retrieve information efficiently.

Examples of Data Stored -

1. User information
2. Passwords
3. Product details
4. Orders
5. Students records

6. Messages

Examples of Databases -

- MySQL
- PostgreSQL
- MongoDB
- Oracle

5. DNS (Domain Name System) is a system that translates human-readable domain names into IP addresses.

Example -

When a user enters google.com, DNS finds the corresponding IP address of Google's server. Without DNS, users would have to remember numerical IP addresses instead of website names.

Date: / /
Page:
6. An IP Address (Internet Protocol Address) is a unique number assigned to every device connected to a network.

Importance -

1. Identifies devices on a network
 2. Enables communication between devices
 3. Helps route internet traffic correctly.
- ex - 192.168.1.1

7. Client - Server Architecture is a model where a client requests services and a server provides them.

Communication Process:

1. Client sends a request.
 2. Server receives the request.
 3. Server processes it.
 4. Server sends a response.
 5. Client displays the result.
- ex - A web browser requesting a webpage from a server.

8. API (Application Programming Interface) is a set of rules that allows different software applications to communicate with each other.

Role of APIs:

- Exchange data between systems.
- Connect frontend and backend
- Integrate third-party services

ex - A weather application using a weather API to fetch weather information.

9. HTTP methods define the actions performed on resources.

1. GET -

→ used to retrieve data from a server.
ex - Viewing a product list.

2. POST -

→ used to send new data to a server.
ex - creating a new user account.

3. PUT -

→ used to update existing data.

ex - updating a user profile.

4. DELETE

→ used to remove data.

ex - deleting a product from a cart.

10. Developer Tools are browser tools that helps developers inspect, debug and analyze web applications.

Three Important Tabs -

Elements Tab - used to inspect and modify HTML and CSS.

Network Tab - shows all network requests made by the website.

Console Tab - displays errors, warnings, and JavaScript output.

Part B -

11. Full Stack Architecture Diagram

User - person using the website or applications.

↓
Frontend - user interface visible in the browser.

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Backend - processes requests and business logic

↓
Database - stores and retrieves application data.

12. User → Browser → DNS → Server → Database →
Server → Browser → Website Displayed

Explanation of Each Step -

1. User enters a website URL.
2. Browser sends a request for the website.
3. DNS converts the domain name into an IP address.
4. Server receives the request.
5. Database provides required data to the server.
6. Server processes the data and prepares a response.
7. Browser receives the response.
8. Website Displayed on the user's screen.

Part C -13. Student Registration SystemFrontend

- Student enters name, email and other details into a form.
- Form validates required fields.

Backend

- receives form data.
- checks if the information is valid.
- processes the registration request.

Database

- stores the student's information permanently.
- allows future retrieval of student records.

14. Amazon Search Scenario
When a user searches for "laptop" on Amazon:

1. The user enters "laptop" in the search box using Browser.
2. The browser sends the search request to the server.
3. The server processes the request.
4. The server queries the database for products related to "laptop".
5. The database returns matching products.

6. The server prepares a Response containing product information.
7. The response is sent back to the browser.
8. The browser displays the laptop search results to the user.

15. Website Observation Activity

website Name: Google

Observation:-

Elements Tab - The page contains HTML elements such as forms, buttons and images.

Network Tab - Multiple requests are made to load images, scripts and stylesheets.

Console Tab - JavaScript messages and browser logs are displayed.

16. When a user enters a URL in a browser, the browser contacts the DNS servers to find the website's IP address.

Using this IP address, the browser sends a request to the web server.

The server processes the request and may retrieve data from a database.

After processing, the server sends a response containing HTML, CSS, Javascript and data. The browser sends this information and displays the webpage on the screen.

17 Frontend	Backend	Database
The Amazon pages that users see, such as product listings, search bars, and shopping carts.	Processes searches, handles orders, manages user accounts and performs business logic.	stores product information, customer details, order history and inventory data.

Example - When a customer searches for a laptop on Amazon, the frontend collects the search term, the backend processes the request and the database provides matching laptop records. The backend sends the results back to the frontend, which displays them to the customer.